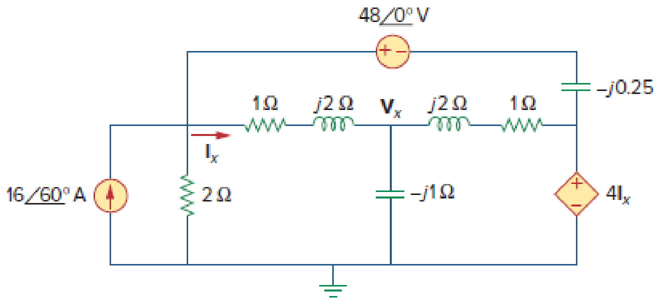
Chapter 10, Problem 14PP

Problem

Obtain **V**x and **I**x in the circuit depicted in Fig. 10.40.

Figure 10.40:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.30 in the textbook.

Consider the value of the angular frequency.

$$\omega = 1 \text{ rad/s}$$

Calculate the values of the inductor, L_1.

$$\begin{aligned} X_{L1} &= j2 \\ j\omega L_1 &= j2 \\ j(1)L_1 &= j2 \\ L_1 &= 2 \text{ H} \end{aligned}$$

Calculate the values of the inductor, L_2.

$$\begin{aligned} X_{L2} &= j2 \\ j\omega L_2 &= j2 \\ j(1)L_2 &= j2 \\ L_2 &= 2 \text{ H} \end{aligned}$$

Step 2 of 4

Calculate the values of the capacitor, C_1.

$$X_{C1} = -j$$
$$\frac{-j\omega}{C_1} = -j$$
$$\frac{-j(1)}{C_1} = -j$$
$$C_1 = 1\text{ F}$$

Calculate the values of the capacitor, C_2.

$$X_{C2} = -j$$
$$\frac{-j\omega}{C_2} = -j0.25$$
$$\frac{-j(1)}{C_2} = -\frac{j}{4}$$
$$C_2 = 4\text{ F}$$

Calculate the frequency, f.

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{1}{2\pi} \\ &= 0.15915 \end{aligned}$$

Step 3 of 4

Draw the circuit in PSPICE schematics.

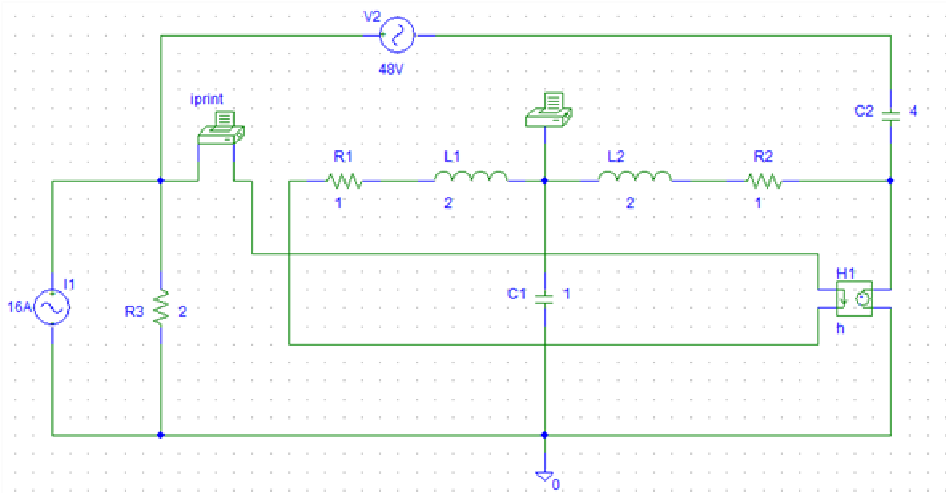


Figure 1

Step 4 of 4

The magnitude and phase of $\mathbf{V}_x$ and $\mathbf{I}_x$ are required. So, set the attributers of I PRINT and V PRINT1 each to AC = yes, MES = yes, PHASE = yes.

Go to 'Analysis' menu open setup / AC sweep and enter total pts as 1, start frequency as 0.15915 and Final frequency as 0.15915

Save the Schematic and simulate it by selecting analysis/simulate. The output file includes the source frequency in addition to the attributes checked for the pseudo components IPRINT and VPRINT1 as follows:

The outputs of the circuit are,

FREQ VM($N_0003) VP($N_0003)

1.592E-01 3.409E+01 4.955E+01

FREQ IM(V_PRINT10) IP(V_PRINT10)

1.592E-01 8.945E+00 1.627E +02

Therefore, the value of the voltage V_x is $\boxed{34.09\angle 49.55^\circ \text{ V}}$ and the current, I_x is $\boxed{8.945\angle 162.7^\circ \text{ A}}$.